

Principal stratification

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Post-treatment variable

- Post-treatment variable (S): variables that are affected by the treatment (Z) and possibly affect the outcome (Y)
- Studies with post-treatment variables include
 - clinical trials where the treatment receipt may be different from the treatment assignment
 - studies with long follow-up, where the dropout of the units is a post-treatment variable
 - clinical trials, where the outcome may be truncated by death
 - studies with time-consuming or costly outcomes, where the surrogate (biomarker) is a post-treatment variable
- Adjusting for the post-treatment variable is necessary
 - post-treatment variable encodes characteristics of the unit as well as of the treatment
 - ACE is not well defined, or cannot answer the question of interest

Post-treatment variable adjustment

- Noncompliance in randomized trials
 - Z_i : treatment assignment; S_i : treatment receipt
 - intention-to-treat (ITT) effect: effect of $Z \neq$ treatment effect
- Truncation by death in randomized trials
 - S_i : survival indicator
 - Y_i is well defined only if $S_i = 1 \rightsquigarrow \mathbb{E}\{Y_i(1) - Y_i(0)\}$ not well defined
 - naive method: $\mathbb{E}(Y_i \mid Z_i = 1, S_i = 1) - \mathbb{E}(Y_i \mid Z_i = 0, S_i = 1)$
- Surrogate
 - ACE provides no information about surrogate S
 - criterion for a good surrogate: $Z \perp\!\!\!\perp Y \mid S$?

Issue of adjusting for observed post-treatment variable

- Potential values: $Y_i(z)$ and $S_i(z)$
- Observed variables: $S_i = S_i(Z_i)$ and $Y_i = Y_i(Z_i)$
- Randomized experiments: $\{Y_i(1), Y_i(0), S_i(1), S_i(0)\} \perp\!\!\!\perp Z_i$
- A causal effect is defined to be the comparison between **potential outcomes** for the **same units**
 - Comparison of $\{Y_i(1), i \in \text{set}_1\}$ and $\{Y_i(0), i \in \text{set}_0\}$
 - Two sets should be identical: $\text{set}_1 = \text{set}_0$
 - ACE: $\mathbb{E}(Y_i \mid Z_i = 1) - \mathbb{E}(Y_i \mid Z_i = 0) = \mathbb{E}\{Y_i(1)\} - \mathbb{E}\{Y_i(0)\}$

Issue of adjusting for observed post-treatment variable

- Comparison of $\mathbb{E}(Y_i \mid Z_i = 1, S_i = 1)$ and $\mathbb{E}(Y_i \mid Z_i = 0, S_i = 1)$
 - $\mathbb{E}(Y_i \mid Z_i = z, S_i = 1) = \mathbb{E}\{Y_i(z) \mid Z_i = z, S_i(z) = 1\} = \mathbb{E}\{Y_i(z) \mid S_i(z) = 1\}$
 - comparison of sets $\{i : S_i(1) = 1\}$ and $\{i : S_i(0) = 1\}$
- Adjusting for the observed post-treatment variable does **NOT** yield valid causal quantities

Issue of adjusting for observed post-treatment variable

- Treatment: Z_i ; survival indicator: S_i ; outcome: Y_i
- Treatment has no effect on outcome: $Y_i(1) = Y_i(0)$ for all i
- Individuals with $S_i(0) = 1$ are healthier than those with $S_i(1) = 1$
 - Larger $Y_i(z)$ for individuals with $S_i(0) = 1$
- Analysis using the survived individuals

$$\begin{aligned} & \mathbb{E}(Y_i \mid Z_i = 1, S_i = 1) - \mathbb{E}(Y_i \mid Z_i = 0, S_i = 1) \\ = & \mathbb{E}\{Y_i(1) \mid S_i(1) = 1\} - \mathbb{E}\{Y_i(0) \mid S_i(0) = 1\} \\ = & \mathbb{E}\{Y_i(0) \mid S_i(1) = 1\} - \mathbb{E}\{Y_i(0) \mid S_i(0) = 1\} \\ < & 0 \end{aligned}$$

The hazard of hazard ratio (Hernán, 2010)

- Treatment Z_i and time-to-event outcome Y_i
- Hazard: the event rate at time t conditional on survival until time t or later

$$\lim_{\Delta t \rightarrow 0} \frac{\text{pr}(t \leq Y < t + \Delta t \mid Y \geq t)}{\Delta t}$$

- Survival analysis assumes models for hazard, e.g., Cox model, additive hazard model
- Hazard ratio between the treatment and control compares

$$\lim_{\Delta t \rightarrow 0} \frac{\text{pr}(t \leq Y(1) < t + \Delta t \mid Y(1) \geq t)}{\Delta t}, \quad \lim_{\Delta t \rightarrow 0} \frac{\text{pr}(t \leq Y(0) < t + \Delta t \mid Y(0) \geq t)}{\Delta t}$$

which compares the populations $\{i : Y(1) \geq t\}$ and $\{i : Y(0) \geq t\}$

- Hazard ratio has a built-in selection bias

Principal stratification

- Principal stratification: stratification based on the joint potential values $\{S_i(1), S_i(0)\}$, denoted by U_i
 - U_i is an unobserved variable
 - U_i is unaffected by the treatment, similar to a covariate
 - binary $S_i \rightsquigarrow$ four valued U_i
 - principal effect: comparison of $\{Y_i(1) : U_i = u\}$ and $\{Y_i(0) : U_i = u\}$
- Any principal effect is a causal effect
 - principal causal effect: $ACE_u = \mathbb{E}\{Y_i(1) - Y_i(0) \mid U_i = u\}$
 - scientific meanings differ in different applications

Non-compliance

- Treatment assignment Z ; treatment receipt S_i
- Principal average causal effect: $ACE_u = \mathbb{E}\{Y_i(1) - Y_i(0) \mid U_i = u\}$,
 $u = a, c, n, d$

$$ACE = \sum_{u=a,c,n,d} ACE_u \cdot \text{pr}(U_i = u)$$

- Identification assumptions
 - monotonicity: $\text{pr}(U_i = d) = 0$
 - exclusion restriction: $ACE_u = 0$ for $u = a, n$
- $CACE = ACE_Y / ACE_S$

Truncation by death

- Example 1: the effect of treatment (Z) on quality of life (Y) $\rightsquigarrow$ some patients may die before the outcome is observed
- Example 2: the effect of job training on hourly wage $\rightsquigarrow$ some subjects may be unemployed
- In example 1, S_i is the survival status and in example 2, S_i is the employment status
- Traditional method treats the truncation by death problem as a standard missing data problem (censoring by death)
 - Heckman selection model: models for $\mathbb{E}(Y_i | Z_i, \mathbf{X}_i)$ and $\mathbb{E}(S_i | \mathbf{X}_i)$
 - assumes that the outcome is well-defined for all units

Principal stratification for truncation by death problem

- Four principal strata:
 - always survivors: $(S_i(1) = 1, S_i(0) = 1) \rightsquigarrow Y_i$ always well defined
 - $(S_i(1) = 1, S_i(0) = 0) \rightsquigarrow Y_i$ not well defined when $Z_i = 0$
 - $(S_i(1) = 0, S_i(0) = 1) \rightsquigarrow Y_i$ not well defined when $Z_i = 1$
 - $(S_i(1) = 0, S_i(0) = 0) \rightsquigarrow Y_i$ never well defined
- Comparison of $Y_i(1)$ and $Y_i(0)$ is valid only for always survivors
- Survivor average causal effect (Rubin, 2006)

$$\text{SACE} = \mathbb{E}\{Y_i(1) - Y_i(0) \mid S_i(1) = 1, S_i(0) = 1\}$$

- Other principal causal effects are not well defined

Surrogate

- Outcomes may be time-consuming or costly to measure
 - Develop of measurement tool and training of people
 - Long term outcomes: long-term survival in clinical trials and long-run return to early life interventions
- Surrogate: a proxy of outcome that is easy to measure
 - should be strongly **associated** with outcome? $\rightsquigarrow$ large $\text{cor}(S_i, Y_i)$
 - should be on the **causal pathway** from treatment to outcome?
 $Z_i \rightarrow S_i \rightarrow Y_i$
- Examples of surrogates
 - CD4 cell count as surrogate for the survival status of HIV patients
 - short-term survival as surrogate for long-term survival
 - kindergarten test scores as surrogate for long-run return

Catastrophic consequence using an invalid surrogate

- Ventricular arrhythmia as surrogate for death related to cardiac complications
 - VA is associated with a fourfold increase in the risk for death
 - it was hypothesized that suppression of ventricular arrhythmias would reduce the rate of death
- Drugs found to suppress arrhythmia were approved by the FDA
 - more than 200 000 persons per year took these drugs
- Follow up trials showed that the drugs increased mortality
 - tens of thousands of patients died in America's worst drug disaster
 - the casualties were estimated to approach levels close to those of the war in Vietnam

Prentice's criteria of a statistical surrogate

- Treatment Z_i ; surrogate S_i ; outcome Y_i
- S_i is a **statistical surrogate** for a comparison of the effect of $Z_i = 1$ versus $Z_i = 0$ on Y_i , if $Z_i \perp\!\!\!\perp Y_i \mid S_i$ (Prentice 1989)

$$Z_i \longrightarrow S_i \longrightarrow Y_i \implies Z_i \perp\!\!\!\perp Y_i \mid S_i, S_i \not\perp\!\!\!\perp Y_i$$

$$\begin{aligned} & \mathbb{E}\{Y_i(1) - Y_i(0)\} \\ = & \mathbb{E}\{\mathbb{E}(Y_i \mid S_i, Z_i = 1) \mid Z_i = 1\} - \mathbb{E}\{\mathbb{E}(Y_i \mid S_i, Z_i = 0) \mid Z_i = 0\} \\ = & \mathbb{E}\{\mathbb{E}(Y_i \mid S_i) \mid Z_i = 1\} - \mathbb{E}\{\mathbb{E}(Y_i \mid S_i) \mid Z_i = 0\} \end{aligned}$$

- $\mathbb{E}\{h(S_i(1)) - h(S_i(0))\} = 0$ implies $\mathbb{E}\{Y_i(1) - Y_i(0)\} = 0$
- there exist units with no causal effect of treatment on the surrogate but experience causal effects of treatment on outcome: $S_i(1) = S_i(0)$ but $Y_i(1) \neq Y_i(0)$

Properties of a good surrogate

- Intuitively, a good surrogate should satisfy:
 - if the treatment does not affect surrogate, it does not affect outcome
 - if the treatment affects surrogate, it affects outcome
- Formally, a good surrogate should satisfy:
 - causal necessity (Frangakis and Rubin, 2002)

$$S_i(1) = S_i(0) \implies Y_i(1) = Y_i(0)$$

- causal sufficiency (Gilbert and Hudgens, 2008)

$$S_i(1) \neq S_i(0) \implies Y_i(1) \neq Y_i(0)$$

Principal stratification and principal causal effects

- A simple example: treatment comparison for AIDS patients, where CD4 is the candidate surrogate
- Four principal strata defined by $\{S_i(1), S_i(0)\}$

$$U_i \equiv \{S_i(1), S_i(0)\} \in \{HH, HL, LH, LL\}$$

- Principal average causal effect

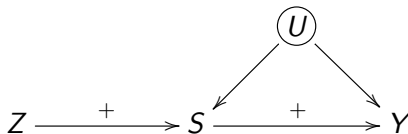
$$ACE_u = \mathbb{E}\{Y_i(1) - Y_i(0) \mid U_i = u\}$$

- Criteria for a principal surrogate
 - causal necessity: $ACE_{HH} = ACE_{LL} = 0$
 - causal sufficiency: $ACE_{HL} \neq 0$ and $ACE_{LH} \neq 0$

Statistical surrogate vs. principal surrogate

- Statistical surrogate: $Z \perp\!\!\!\perp Y \mid S$
principal surrogate: conditions on ACE_u
 - If the post-treatment variable S_i is a principal surrogate, then it is not generally a statistical surrogate
 - If the post-treatment variable S_i is a statistical surrogate, then it is not generally a principal surrogate
- Evaluation
 - statistical surrogate: direct estimation from the observed data
 - principal surrogate: requires the identification and estimation of principal causal effects

Can we avoid the disaster using these criteria?



- Surrogate “paradox” (Chen, Geng and Jia 2007, Ju and Geng 2010)
- For a principal or statistical surrogate, it is still possible that
 - **Positive** causal effect of Z on S : $ACE(Z \rightarrow S) > 0$
 - **Positive** causal effect of S on Y : $ACE(S \rightarrow Y) > 0$ and $Cor(S, Y) > 0$
 - **Negative** causal effect of Z on Y : $ACE(Z \rightarrow Y) < 0$
- It is possible that drugs are **beneficial** for VA but **increase mortality**

Principal surrogates may suffer from surrogate paradox

- Proportions of principal strata, $\pi_u = \text{pr}(U_i = u)$:
 $(\pi_{HH}, \pi_{HL}, \pi_{LH}, \pi_{LL}) = (0.2, 0.4, 0.2, 0.2)$

$$\begin{aligned}\text{ACE}(Z \rightarrow S) &= \text{pr}\{S_i(1) = 1\} - \text{pr}\{S_i(0) = 1\} \\ &= (\pi_{HL} + \pi_{HH}) - (\pi_{HH} + \pi_{LH}) = \pi_{HL} - \pi_{LH} = 0.2 > 0\end{aligned}$$

- Principal causal effects

$$\text{ACE}_{HH} = \text{ACE}_{LL} = 0, \quad \text{ACE}_{HL} = 0.1 > 0, \quad \text{ACE}_{LH} = -0.3 < 0$$

- However, the ACE on the outcome is negative

$$\text{ACE}(Z \rightarrow Y) = \sum_u \pi_u \text{ACE}_u = 0.4 \times 0.1 + 0.2 \times (-0.3) = -0.02 < 0$$

New criteria: when a surrogate satisfies causal necessity

- $\text{ACE}(Z \rightarrow S) > 0$ should imply $\text{ACE}(Z \rightarrow Y) > 0$

$$\text{ACE}(Z \rightarrow Y) = \sum_u \text{ACE}_u \cdot \text{pr}(U_i = u)$$

- If $\text{ACE}_{HL} + \text{ACE}_{LH} \geq 0$, then

$$\text{ACE}(Z \rightarrow Y) \geq \text{ACE}(Z \rightarrow S) \times \text{ACE}_{HL}$$

- $\text{ACE}_{HL} > 0 \implies$ lower bound of $\text{ACE}(Z \rightarrow Y)$ and $\text{ACE}(Z \rightarrow S)$ have the same sign
- New criteria for avoiding surrogate paradox
 - causal necessity: $\text{ACE}_{HH} = \text{ACE}_{LL} = 0$
 - $\text{ACE}_{HL} > 0$ and $\text{ACE}_{HL} + \text{ACE}_{LH} \geq 0$

Statistical inference and its difficulty

- For problems of non-compliance, truncation by death, and surrogate evaluation, we need to identify and estimate principal causal effects
 - we focus on the randomized experiment setting:
 $Z_i \perp\!\!\!\perp (Y_i(z), S_i(1), S_i(0))$
 - key to identification: express ACE_u in terms of the observed distribution (Z, S, Y, X)

- Under randomization of Z

$$ACE_u = \mathbb{E}(Y_i = 1 \mid Z_i = 1, U_i) - \mathbb{E}(Y_i = 1 \mid Z_i = 0, U_i)$$

- difficulty: $S_i(1)$ and $S_i(0)$ are not simultaneously observed $\rightsquigarrow U_i$ is latent
- In noncompliance problems, we can identify CACE under exclusion restriction and monotonicity

Partial identification for truncation-by-death problem

- Treatment Z ; survival status S_i ; outcome Y_i
- SACE: $ACE_{11} = \mathbb{E}\{Y_i(1) - Y_i(0) \mid S_i(1) = 1, S_i(0) = 1\}$
- Exclusion restriction implies $SACE = 0 \rightsquigarrow$ cannot be invoked
 - without exclusion restriction, SACE is not identifiable
 - alternative identification assumptions?
- Partial identification $\rightsquigarrow$ bounds on SACE
 - find all the possible values of SACE that are compatible with the observed data

Bounds for survivor average causal effect

- Randomization and monotonicity hold
- Proportions of principal strata:
 $\pi_{00} = \text{pr}(S_i = 0 \mid Z_i = 1)$, $\pi_{11} = \text{pr}(S_i = 1 \mid Z_i = 0)$, and
 $\pi_{10} = 1 - \pi_{00} - \pi_{11} = \text{pr}(S_i = 1 \mid Z_i = 1) - \text{pr}(S_i = 1 \mid Z_i = 0)$
- $S_i(1) = 1, S_i(0) = 1 \iff S_i(0) = 1$
 $\mathbb{E}\{Y_i(0) \mid S_i(1) = 1, S_i(0) = 1\} = \mathbb{E}\{Y_i(0) \mid S_i(0) = 1\} = \mathbb{E}(Y_i \mid Z_i = 0, S_i = 1)$
- Key: bounds on $\mathbb{E}\{Y_i(1) \mid S_i(1) = 1, S_i(0) = 1\}$

Bounds on survivor average causal effect

- The observed stratum ($Z_i = 1, S_i = 1$) is a mixture of two strata $U_i = 11$ and $U_i = 10$

$$\begin{aligned}\mathbb{E}(Y_i = 1 \mid Z_i = 1, S_i = 1) &= \mathbb{E}\{Y_i(1) = 1 \mid U = 11/10\} \\&= \mathbb{E}\{Y_i(1) \mid U = 11\}\text{pr}(U = 11 \mid U = 11/10) \\&\quad + \mathbb{E}\{Y_i(1) \mid U = 10\}\text{pr}(U = 10 \mid U = 11/10) \\&= \frac{\pi_{11}}{\pi_{11} + \pi_{10}} \mathbb{E}\{Y_i(1) \mid S_i(1) = 1, S_i(0) = 1\} \\&\quad + \frac{\pi_{10}}{\pi_{11} + \pi_{10}} \mathbb{E}\{Y_i(1) \mid S_i(1) = 1, S_i(0) = 0\}\end{aligned}$$

- Y is bounded in $[l, u] \rightsquigarrow$ bounds on $\mathbb{E}\{Y_i(1) \mid S_i(1) = 1, S_i(0) = 1\}$

$$\begin{aligned}\text{Upper} &= \frac{(\pi_{11} + \pi_{10})\mathbb{E}(Y_i = 1 \mid Z_i = 1, S_i = 1) - l \cdot \pi_{10}}{\pi_{11}} \\ \text{Lower} &= \frac{(\pi_{11} + \pi_{10})\mathbb{E}(Y_i = 1 \mid Z_i = 1, S_i = 1) - u \cdot \pi_{10}}{\pi_{11}}\end{aligned}$$

- Statistical inference is hard; even harder without monotonicity

Acute respiratory distress syndrome network study

$Z = 1$				$Z = 0$			
	$Y = 1$	$Y = 0$	total		$Y = 1$	$Y = 0$	total
$S = 1$	54	268	302	$S = 1$	59	218	277
$S = 0$	*	*	109	$S = 0$	*	*	152

- 861 patients with lung injury and acute respiratory distress syndrome were randomized to receive mechanical ventilation with either lower tidal volumes ($Z_i = 1$) or traditional tidal volumes ($Z_i = 0$)
- Outcome: breathe without assistance by day 28 (1 for not)

Acute respiratory distress syndrome network study

$Z = 1$				$Z = 0$			
	$Y = 1$	$Y = 0$	total		$Y = 1$	$Y = 0$	total
$S = 1$	54	268	302	$S = 1$	59	218	277
$S = 0$	*	*	109	$S = 0$	*	*	152

- Proportions of principal strata

$$\hat{\pi}_{11} = \frac{277}{277 + 152} = 0.646, \quad \hat{\pi}_{00} = \frac{109}{109 + 302} = 0.265, \quad \hat{\pi}_{10} = 0.089$$

- Sample means of the outcome for survived patients

$$\hat{\mathbb{E}}(Y_i | Z_i = 1, S_i = 1) = \frac{54}{302} = 0.179, \quad \hat{\mathbb{E}}(Y_i | Z_i = 0, S_i = 1) = \frac{59}{277} = 0.213$$

- Bounds on $\mathbb{E}\{Y_i(1) | S_i(1) = 1, S_i(0) = 1\}$: [0.065, 0.203]
- Bounds on SACE: [-0.147, -0.010]

Principal ignorability

- An alternative set of identification assumptions without assuming exclusion restriction
 - truncation by death, surrogate evaluation, and etc.
 - similar idea as the treatment ignorability assumption for identifying ACE in observational studies
- Treatment ignorability $(Z_i \perp\!\!\!\perp Y_i(z) \mid \mathbf{X}_i) \rightsquigarrow$
 $\mathbb{E}\{Y_i(z) \mid Z_i = 1, \mathbf{X}_i\} = \mathbb{E}\{Y_i(z) \mid Z_i = 0, \mathbf{X}_i\}$
- Principal ignorability $Y_i(z) \perp\!\!\!\perp U_i \mid (Z_i = z, S_i = s, \mathbf{X}_i)$
 - expected potential outcome is the same across different principal strata within each observed stratum
 - implies $\mathbb{E}\{Y_i(1) \mid U_i = 11, Z_i = 1, S_i = 1, \mathbf{X}_i\} = \mathbb{E}\{Y_i(1) \mid U_i = 10, Z_i = 1, S_i = 1, \mathbf{X}_i\}$

Identification assumptions

- Randomization: $Z_i \perp\!\!\!\perp \{Y_i(1), Y_i(0), S_i(1), S_i(0)\}$
- Monotonicity: $S_i(1) \geq S_i(0)$
 - principal score
$$\begin{aligned}\text{pr}(U_i = 11 \mid \mathbf{X}_i) &= \text{pr}(S_i = 1 \mid Z_i = 0, \mathbf{X}_i), \\ \text{pr}(U_i = 00 \mid \mathbf{X}_i) &= \text{pr}(S_i = 0 \mid Z_i = 1, \mathbf{X}_i) \\ \text{pr}(U_i = 10 \mid \mathbf{X}_i) &= \text{pr}(S_i = 1 \mid Z_i = 1, \mathbf{X}_i) - \text{pr}(S_i = 1 \mid Z_i = 0, \mathbf{X}_i)\end{aligned}$$
- Principal ignorability: $Y_i(z) \perp\!\!\!\perp U_i \mid (Z_i = z, S_i = s, \mathbf{X}_i)$
 - $\mathbb{E}\{Y_i(1) \mid U_i = 11, Z_i = 1, S_i = 1, \mathbf{X}_i\} = \mathbb{E}\{Y_i(1) \mid Z_i = 1, S_i = 1, \mathbf{X}_i\}$
 - $\mathbb{E}\{Y_i(1) \mid U_i = 10, Z_i = 1, S_i = 1, \mathbf{X}_i\} = \mathbb{E}\{Y_i(1) \mid Z_i = 1, S_i = 1, \mathbf{X}_i\}$

Weighting method

Theorem

$$ACE_{10} = \mathbb{E}\{w_{1,10}(\mathbf{X})Y \mid Z = 1, S = 1\} - \mathbb{E}\{w_{0,10}(\mathbf{X})Y \mid Z = 0, S = 0\}$$

$$ACE_{00} = \mathbb{E}\{Y \mid Z = 1, S = 0\} - \mathbb{E}\{w_{0,00}(\mathbf{X})Y \mid Z = 0, S = 0\}$$

$$ACE_{11} = \mathbb{E}\{w_{1,11}(\mathbf{X})Y \mid Z = 1, S = 1\} - \mathbb{E}\{Y \mid Z = 0, S = 1\}$$

$$\pi_u(\mathbf{X}) = \text{pr}(U_i = u \mid \mathbf{X}) \quad \pi_u = \mathbb{E}\{\pi_u(\mathbf{X})\}$$

$$w_{1,10} = \frac{\pi_{10}(\mathbf{X})}{\pi_{10}(\mathbf{X}) + \pi_{11}(\mathbf{X})} \bigg/ \frac{\pi_{10}}{\pi_{10} + \pi_{11}} \quad w_{0,11} = \frac{\pi_{10}(\mathbf{X})}{\pi_{10}(\mathbf{X}) + \pi_{00}(\mathbf{X})} \bigg/ \frac{\pi_{10}}{\pi_{10} + \pi_{00}}$$

$$w_{0,00} = \frac{\pi_{00}(\mathbf{X})}{\pi_{10}(\mathbf{X}) + \pi_{00}(\mathbf{X})} \bigg/ \frac{\pi_{00}}{\pi_{10} + \pi_{00}} \quad w_{1,11} = \frac{\pi_{11}(\mathbf{X})}{\pi_{10}(\mathbf{X}) + \pi_{11}(\mathbf{X})} \bigg/ \frac{\pi_{11}}{\pi_{10} + \pi_{11}}$$

Estimation and inference

- Estimation steps
 - 1 estimate the principal score $\text{pr}(U_i | \mathbf{X}_i)$, e.g., logistic model
 - 2 plug the estimated principal score in the weighting formula
- Variance and confidence interval: bootstrap
- A unification in observational studies (Jiang et al., 2022)
 - treatment ignorability, principal ignorability, monotonicity
 - triple robustness: propensity score, principal score, outcome model

Return to schooling

- IV Z : growing up near a four-year college; S : education beyond high school: outcome Y : wage in 1976
- Target: the effect of education on wage
- Assumptions
 - monotonicity: $S_i(1) \geq S_i(0)$
 - principal ignorability given race, age, living with parents, living areas in the past
 - no exclusion restriction
- Results
 - $ACE_{00} = 0.02$ (CI $(-0.03, 0.07)$) and $ACE_{11} = 0.01$ (CI $(-0.05, 0.07)$)
 - $ACE_{10} = 0.10$ (CI $(-0.01, 0.23)$)

Comparison of different identification strategies

- A1: $Z_i \perp\!\!\!\perp \{Y_i(1), Y_i(0), S_i(1), S_i(0)\}$
A2: $Z_i \perp\!\!\!\perp \{Y_i(1), Y_i(0), S_i(1), S_i(0)\} \mid \mathbf{X}_i$
- B1: monotonicity and exclusion restriction
B2: monotonicity and principal ignorability
- A1+ B1: non-compliance in randomized trials, model free
- A2+ B1: non-compliance in observational studies, model for Z_i or (S_i, Y_i)
- A1+B2: randomized trials without exclusion restriction, model for S_i
- A2+B2: observational studies without exclusion restriction, models for two of Z_i , S_i , and Y_i
- Other strategies: additional information (e.g., auxiliary independence), likelihood based inference, Bayesian analysis

Summary

- Post-treatment variable S_i : affected by the treatment
 - naive adjustment for the observed variable S_i does not yield a valid causal quantity
 - principal stratification defined by $S_i(1)$ and $S_i(0)$
 - application to non-compliance, truncation-by-death, and surrogate evaluation problems
- Various identification strategies
 - monotonicity and exclusion restriction
 - monotonicity and principal ignorability
- Extension
 - identification without monotonicity
 - discrete and continuous S

Suggested readings

- Principal stratification
 - Frangakis and Rubin. 2002. “Principal stratification in causal inference”
- Truncation by death
 - Rubin. 2006. “Causal Inference Through Potential Outcomes and Principal Stratification: Application to Studies with “Censoring” Due to Death”
- Surrogate evaluation
 - VandeWeele. 2013. “Surrogate measures and consistent surrogates”
 - Jiang, et al. 2016. “Principal causal effect identification and surrogate endpoint evaluation by multiple trial”
- Identification under principal ignorability
 - Ding and Lu. 2017. “Principal stratification analysis using principal scores”
 - Jiang, Yang and Ding 2022. “Multiply robust estimation of causal effects under principal ignorability”